A Reproduction Audit of *Deep Autoencoder-Based Anomaly Detection of Electricity Theft Cyberattacks in Smart Grids*

ATK Evidence

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Abstract

This report tests whether the principal numerical results reported by Takiddin et al. can be reproduced from the named datasets and the experimental procedure described in the paper. The primary track preserves the printed method, including questionable operations; material omissions and contradictions are tested as predeclared branches; statistically corrected controls are reported separately. This document is a live report scaffold. One full compact FC-SAE interpretation has completed; no paper-level reproduction verdict has yet been issued.

1 Scope and testable claim

The target is the complete result pattern in Tables II–V, not an isolated metric or a favorable random seed. A paper-consistent reproduction must recover the joint metric pattern within predeclared tolerances across repeated runs. Failure is bounded to the finite documented implementation, hyperparameter, seed, split, and compute space; it is not a claim of logical impossibility or author intent.

2 Source protocol reconstruction

The paper uses two different data paths. SGCC supplies real benign and malicious labels. ISET, drawn from CER, supplies benign half-hourly profiles, from which six malicious variants are synthesized using Equations (1)–(6). Anomaly detectors are trained on benign data only, whereas supervised models use benign and malicious classes. The printed anomaly protocol applies ADASYN to the test set; the supervised protocol applies ADASYN before splitting.

In the frozen ISET construction, the anomaly test population before ADASYN has 750,770 benign profiles and 13,507,740 malicious profiles. The selected exact imbalanced-learn/scikit-learn default therefore queries 750,770 minority rows against all 14,258,510 test rows in its first nearest-neighbor pass, or roughly 10.7×10^{12} profile-pair distances, followed by a minority-only search and generation of approximately 12.8 million synthetic benign profiles. A bounded full attempt consumed 4,724.52 seconds wall time and 33,665.16 CPU-seconds without completing the first query. This is an operational result for the declared default implementation, not a proof that ADASYN is impossible in principle: exact GPU search, approximate neighbors, or an unreported subsample may have different cost, and the paper specifies none of them. A same-machine 16-core benchmark of 250 queries against all 14,258,510 rows processed 3,564,627,500 distance pairs in 16.12 seconds.

Linear extrapolation estimates 13.45 hours for the first full search and 0.71 hours for the minority-only search, or 14.16 wall-hours before synthesis and serialization. This makes the selected CPU default expensive but plainly possible as an overnight job; it provides no basis for claiming that the authors could not have run ADASYN.

3 Reproduction methods

3.1 Data provenance

Every input is identified by source, access conditions, version, byte size, and checksum. Raw datasets and the source paper remain local and are not redistributed.

3.2 Paper-consistent branches

Each material omission or contradiction receives a stable branch identifier before eligible results are examined. Non-executable printed expressions are preserved as findings and receive only explicitly labeled minimal repairs.

3.3 From impossible wording to executable evidence

When a printed operation cannot exist or execute, the audit does not silently repair it and does not stop at criticism. First, the exact wording and source locator are preserved as a literal non-executable outcome. Second, the smallest reasonable executable repair and every other materially defensible reading are predeclared under separate interpretation identifiers. Third, every such repair is executed over all declared seeds. The results section and public study site will place the reported target, literal failure, every repair's configuration, failures, timings, and metric distribution side by side. A repair that matches the paper is reported as plainly as a non-match. No repair is represented as the authors' unobserved implementation. The working hypothesis is non-match, but it cannot determine branch selection, stopping, or display.

3.4 Source failures and planned executable interpretations

Source issue	Literal status	Executable evidence path
SGCC length versus input	Roughly 1,034 daily values are supplied but the model consumes 48 inputs; no mapping is stated.	Separately named crop, window, resampling, and aggregation interpretations; no proxy may be called the literal result.
Attack 3 endpoint	$t_f = t_i - t_l$ puts the final time before the start; hours are also mapped to 48 half-hour positions without a rule.	Preserve the literal failure; execute addition-plus-clipping, addition-plus-wraparound, and valid-endpoint redraw with declared slot mappings.
Customer identity	The text requires unseen test customers but constructs the test population as $B_2 + M$, where M was generated from every customer.	Run all-customer versus held-out-customer attacks and customer-disjoint versus daily-row splits as distinct interpretations.

Source issue	Literal status	Executable evidence path
VAE definition	Equation (9) mixes incompatible distribution variables; variance positivity and reconstruction probability are undefined; two passages give opposite score directions.	Execute finite declared loss reductions, positive variance parameterizations, probability estimators, draw counts, and both printed score directions.
AEA decoder	Algorithm 5 uses the reconstruction before it has been decoded and disagrees with the prose over summation versus concatenation.	Execute repeat-latent, first-step/zero, and autoregressive schedules with both merge readings under separate identifiers.
Search and threshold	Benign-only X_{TR} cannot provide DR/FA or an ROC; the layer loop cannot yield the unequal Table-I widths; "median of IQR of ROC" is not an algorithm.	Preserve the failures; execute direct Table-I replay and predeclared staged-search, validation-population, and deterministic threshold interpretations.
Reported metric identities	Table-II Naive Bayes has DR=PR=75 but F1=77, and Tables II–III admit no common prevalence from their DR/FA/PR rows under the printed formulas.	This is a static source contradiction, not something to repair by changing the target. Experimental rows remain reported separately against the exact published values.
Table-V identity	A fixed model, threshold, and benign set imply one FA, but six FAs are reported per model.	Execute common-model/common-benign, per-attack retraining, per-attack benign resampling, and their combination.

3.5 Corrected controls

Corrected experiments use customer-disjoint partitions, transformations fitted on training data only, untouched validation/test data, and valid validation-selected thresholds. They answer a different question and cannot substitute for the paper-consistent reproduction.

4 Fidelity audit

The source-to-code audit will cover data populations, attack equations, split identity, normalization and ADASYN order, every reported neural layer, training loss, anomaly score, threshold selection, metric definitions, and experiment identity for Tables II–V. Historical runs affected by a material mismatch are retained but excluded from paper-reproduction claims.

5 Results

Exploratory: One full compact `I-ADASYN-NONE-ISET-FC-SAE` attempt has completed. It is the frozen seed-11, batch-512 executable completion without test-set resampling, not the completed printed-ADASYN result and not a confirmatory verdict. The model had 450,448 parameters, trained for 74 epochs with epoch 69 restored, and completed in 53:12 of Slurm wall time.

Table 2: Reported and reproduced full-ISET FC-SAE metrics (percent).

	DR	FA	Balanced ACC	AUC	F1
Reported	81.00	15.00	83.00	81.00	81.00
Reproduced, seed 11	26.18	58.22	33.98	31.04	40.46

The result closely repeats the prior batch-32 sensitivity rather than being rescued by the frozen batch-512 primary completion. Under the common-model, common-benign Table-V interpretation, reproduced FA is exactly 57.9152% for all six attack types, whereas the paper reports 15, 16, 10, 17, 17, and 19%. An artifact-reload score audit gives additional diagnosis. With the paper's "larger reconstruction error is malicious" direction, an oracle threshold selected on the test labels reaches only 50.0019% balanced accuracy (AUC 31.0439%). Reversing the score direction reaches 66.2554% balanced accuracy (AUC 68.9561%), still 16.7446 points below the reported 83%. Moreover, trained reconstruction-error scores correlate at 0.999459 with the scores obtained by reconstructing every profile as zero. This is consistent with a structural tension in the printed method: joint zero-mean/unit-variance standardization produces negative inputs, while Table I's Softmax output is nonnegative and constrained to sum to one. The diagnostic motivates a separately labeled linear-output repair; it does not modify or invalidate the paper-consistent Softmax attempt.

The additional-seed audit remains open. Each non-executable source item will ultimately receive a reported-versus-literal-failure-versus-repair block; no closest repair will be selected in place of the full predeclared family.

5.1 A conditional bound on what test-set ADASYN can repair

For ISET, ADASYN oversamples the benign test class after model fitting. It does not alter the fitted weights or predictions on the preserved malicious rows. Thus, for this trained model, DR remains 26.1807% regardless of the added benign samples. From the paper's own balanced-accuracy definition,

$$\mathrm{ACC} = \frac{\mathrm{DR} + 100 - \mathrm{FA}}{2},$$

even the unattainably favorable bound FA = 0 gives only 63.0903%, below the reported 83%. This is a mathematical impossibility result conditional on this trained model and the paper's ISET ADASYN placement. It is not a proof that every paper-consistent training completion has the same DR, nor evidence of fabrication or author intent.

6 Statistical assessment

The final analysis will report all predeclared seeds and failures, uncertainty intervals, complete metric patterns, and the fraction of eligible branches meeting the frozen reproduction tolerance. No best-seed or best-branch result will stand in for the distribution. The current 63.0903% ceiling is an exact conditional bound, not a substitute for repeated seeds or the finite interpretation envelope.

Table 3: Final reproduction decisions will be populated only from eligible, fingerprinted attempts.

Published table	Reproduction state	Evidence status
Table II	Pending eligible completion	Open
Table III	One no-resampling FC-SAE seed	Exploratory
Table IV	Full-size FC-SAE timing recorded	Exploratory
Table V	Common-model FC-SAE identity recorded	Exploratory

The first exploratory summary uses three frozen seeds without selecting the best. Confirmation will freeze a larger seed set and the finite source-supported interpretation envelope. A branch will be classified as a non-match only when its uncertainty interval remains outside the predeclared ± 2 percentage-point reproduction band. Because the paper reports rounded point estimates without repetitions or uncertainty, a two-sample test against the authors' run distribution is not possible. Conclusions therefore remain bounded to the tested envelope rather than an infinite space of unreported implementations.

7 Limitations

The paper omits hardware, software versions, epochs, batch size, seeds, repetitions, and several data/threshold details. Consequently, any verdict must remain conditional on the finite tested interpretations. Reported training minutes cannot be compared absolutely without the original timing boundary and hardware.

8 Verdict

Not yet issued. The verdict will be frozen only after the eligible reproduction matrix and its statistical assessment are complete.

A Reproducibility materials

Code, configurations, provenance records, checksums, run manifests, and machine-readable summaries are maintained at `https://github.com/fjoad/atk-evidence`. The public method map is intended to make the paper's experimental flow readable before the detailed evidence is examined.